



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Capstone Project

AI-Based Human Drowning Detection in Swimming Pools Using YOLO V6 with Deep Learning

Team no. 07

MLA0403 – Deep Learning

Guided By

Dr.A.M.Arul Raj

Professor

Institute of SIMATS

SIMATS Engineering

Students Name

N Sai Nikhileswar - 192425330

K Lokesh Chandra - 192425358

D Masthan Vali - 192425374

G Venkata Ganesh - 192425381

P Bharath - 192425368

Institute of SIMATS

SIMATS Engineering

Abstract

- AI-based drowning detection using YOLOv6.
- Monitors swimming pools in real time.
- Detects drowning behavior automatically.
- Uses deep learning for accurate detection.
- Sends instant emergency alerts.
- Improves swimmer safety.

Aim and Objective

- Real-time drowning detection
- YOLOv6-based monitoring
- Enhanced swimmer safety
- Detect drowning incidents
- Send instant emergency alerts
- Reduce rescue response time

Introduction

- AI-based drowning detection system
- YOLOv6 for real-time monitoring
- Deep learning for human detection
- Early emergency alert generation
- Faster rescue response
- Enhanced swimming pool safety



Literature Review & Requirement Analysis

S.No	Year of Publication	Author	Title of the Journal / Book	Methodology Used	Algorithm Used	Key Findings
1	2022	Li et al.	AI-Based Drowning Detection in Swimming Pools	CCTV Video Analysis	YOLOv5	Real-time drowning detection with high accuracy.
2	2023	Zhang et al.	Deep Learning for Human Activity Recognition	Video Surveillance	CNN	Improved recognition of drowning behavior.
3	2024	Kumar et al.	Intelligent Pool Safety Monitoring System	Deep Learning-Based Detection	YOLOv6	Faster detection and instant emergency alerts.

Problem Statement

- Drowning incidents occur due to delayed detection.
- Manual monitoring is not always reliable.
- Lack of real-time automated surveillance.
- Delayed alerts increase the risk of fatalities.
- Need for an AI-based YOLOv6 drowning detection system.

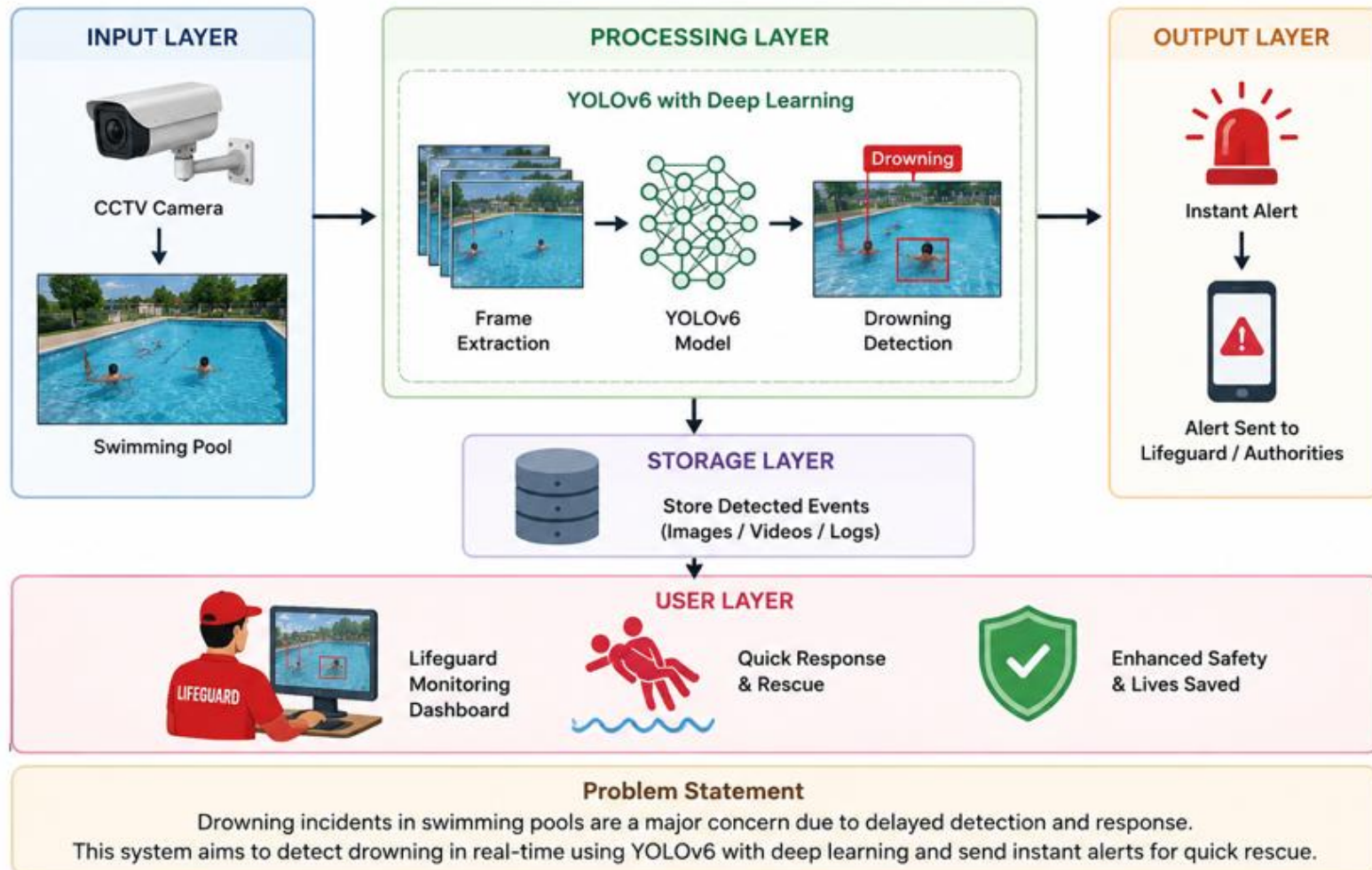
Existing System

- Manual monitoring by lifeguards.
- CCTV surveillance without AI analysis.
- Delayed drowning detection.
- No automatic emergency alerts.
- Higher risk of rescue delays

Proposed System

- AI-based drowning detection using YOLOv6.
- Real-time CCTV video monitoring.
- Automatic human behavior analysis.
- Instant emergency alert generation.
- Faster rescue and improved swimmer safety.

Overall System Architectural Design





Sustainable Development Goal (SDG) Implementation

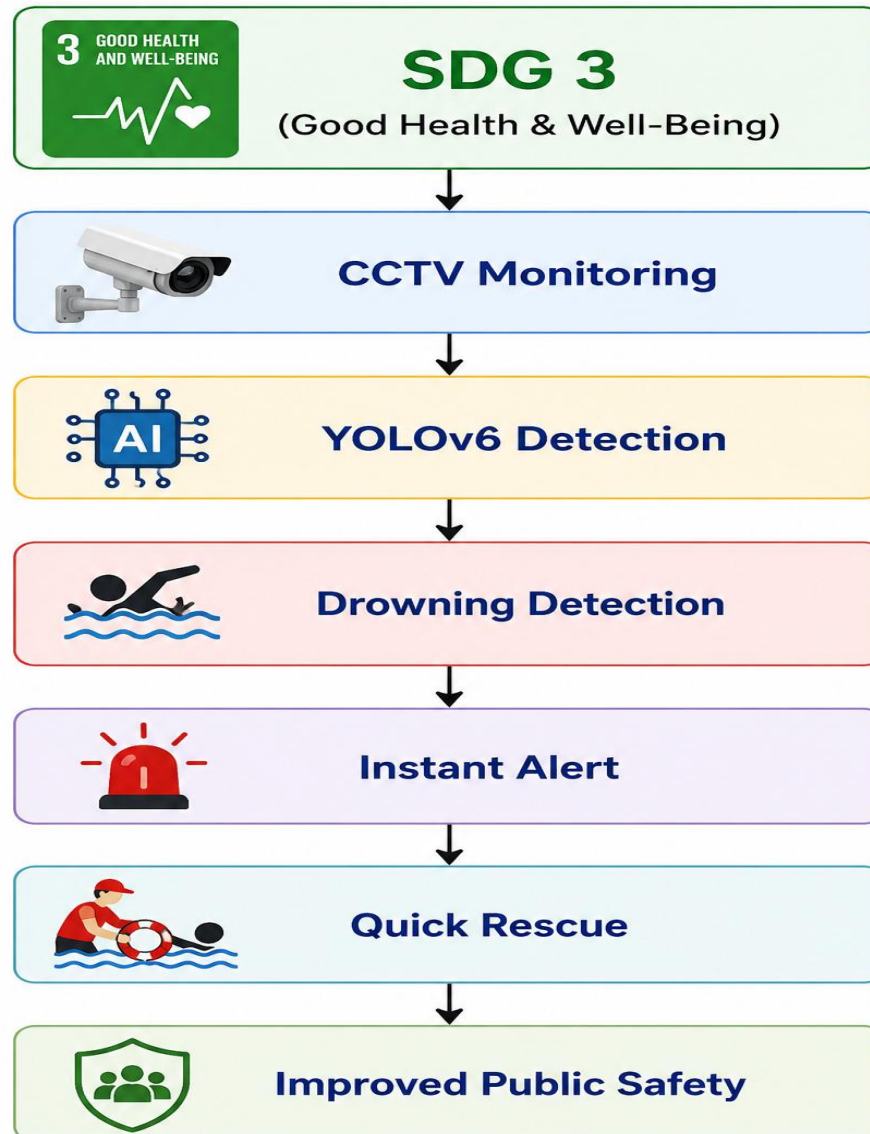


SDG 3 – Good Health and Well-Being

- AI-based drowning detection.
- Real-time CCTV monitoring.
- YOLOv6 human detection.
- Early emergency alerts.
- Faster rescue response.



Simple SDG Flowchart



Proposed Methodology and Algorithm

Algorithm: YOLOv6 (You Only Look Once Version 6)

- Capture live video using CCTV cameras.
- Extract video frames for processing.
- Detect humans using the YOLOv6 model.
- Identify drowning behavior in real time.
- Generate instant alerts for quick rescue.



Design & Module Planning

- Module 1: Video Acquisition
- Module 2: Video Preprocessing
- Module 3: Human Detection
- Module 4: Drowning Detection
- Module 5: Alert and Monitoring

Conclusion

- AI-based drowning detection using YOLOv6.
- Real-time swimming pool monitoring.
- Early detection of drowning incidents.
- Instant alerts for quick rescue.
- Improved swimmer safety and reduced fatalities.

References

- **Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016)** – *You Only Look Once: Unified, Real-Time Object Detection.*
- **Redmon, J., & Farhadi, A. (2018)** – *YOLOv3: An Incremental Improvement.*
- **Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020)** – *YOLOv4: Optimal Speed and Accuracy of Object Detection.*
- **Jocher, G., et al. (2021)** – *YOLOv5 Documentation.*
- **Li, C., Li, L., Jiang, H., et al. (2022)** – *YOLOv6: A Single-Stage Object Detection Framework for Industrial Applications.*
- **He, K., Zhang, X., Ren, S., & Sun, J. (2016)** – *Deep Residual Learning for Image Recognition.*
- **Ren, S., He, K., Girshick, R., & Sun, J. (2015)** – *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.*
- **Simonyan, K., & Zisserman, A. (2015)** – *Very Deep Convolutional Networks for Large-Scale Image Recognition.*



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

THANK YOU